

Solutions to Problem Set 6

1. Expectation, variance and higher moments

- (a) Consider a continuous real random variable X with law characterized by the probability density function

$$f_X(x) = \begin{cases} \frac{c}{(1+x)^5}, & x > 0, \\ 0, & x \leq 0. \end{cases}$$

- Determine the value of c .
 - Calculate the cumulative distribution function of X .
 - Determine the values $\mathbf{E}[X]$ and $\text{Var}[X]$.
Hint: First calculate $\mathbf{E}[1+X]$ and $\mathbf{E}[(1+X)^2]$.
 - What is the largest $n \in \mathbb{N}$ such that $\mathbf{E}[X^n]$ exists? Justify your answer.
- (b) A continuous random variable X has a Γ -distribution with parameters $\alpha > 0$ and $\lambda > 0$, written as $X \sim \Gamma(\alpha, \lambda)$, if its law is characterized by the probability density function

$$f_X(x) = \begin{cases} \frac{1}{\Gamma(\alpha)} \lambda^\alpha x^{\alpha-1} e^{-\lambda x}, & x > 0, \\ 0, & x \leq 0, \end{cases}$$

where $\Gamma(\alpha) = \int_0^\infty x^{\alpha-1} e^{-x} dx$.

- Show that f_X defined above is a density function.
- Show by integration by parts that for all $\alpha > 0$, $\Gamma(\alpha + 1) = \alpha \Gamma(\alpha)$. In particular, since $\Gamma(1) = 1$, conclude that $\Gamma(n) = (n-1)!$ for all $n \in \mathbb{N}$.
- Calculate $\mathbf{E}[X]$ if $X \sim \Gamma(\alpha, \lambda)$, where $\alpha, \lambda > 0$.
- Calculate $\text{Var}[X]$ if $X \sim \Gamma(\alpha, \lambda)$, where $\alpha, \lambda > 0$.

- (a) i. We need $\int_{-\infty}^\infty f(x) dx = 1$, so

$$\int_0^\infty \frac{c}{(1+x)^5} dx = c \left[-\frac{1}{4} \frac{1}{(1+x)^4} \right]_0^\infty = \frac{c}{4} = 1,$$

and therefore, $c = 4$.

- ii. The cumulative distribution function is given by

$$\begin{aligned} F_X(x) &= \mathbf{P}[X \leq x] = \int_{-\infty}^x f_X(t) dt \\ &= \begin{cases} 0, & x < 0, \\ \int_0^x \frac{4}{(1+t)^5} dt, & x \geq 0, \end{cases} \\ &= \begin{cases} 0, & x < 0, \\ \left[-\frac{1}{(1+t)^4} \right]_0^x, & x \geq 0, \end{cases} \\ &= \begin{cases} 0, & x < 0, \\ 1 - \frac{1}{(1+x)^4}, & x \geq 0 \end{cases} \end{aligned}$$

- iii. We calculate

$$\begin{aligned} \mathbf{E}[1+X] &= \int_{-\infty}^\infty (1+x) f_X(x) dx = \int_0^\infty \frac{4}{(1+x)^4} dx = \left[-\frac{4}{3(1+x)^3} \right]_0^\infty = \frac{4}{3}, \\ \mathbf{E}[(1+X)^2] &= \int_{-\infty}^\infty (1+x)^2 f_X(x) dx = \int_0^\infty \frac{4}{(1+x)^3} dx = \left[-\frac{4}{2(1+x)^2} \right]_0^\infty = 2. \end{aligned}$$

It follows that

$$\mathbf{E}[X] = \mathbf{E}[1 + X] - 1 = \frac{1}{3},$$

$$\text{Var}[X] = \text{Var}[1 + X] = \mathbf{E}[(1 + X)^2] - (\mathbf{E}[1 + X])^2 = \frac{2}{9}.$$

Here we used the fact that $\text{Var}[X + b] = \text{Var}[X]$ for every $b \in \mathbb{R}$, see Proposition 7.9, (ii).

iv. We claim that the largest number $n \in \mathbb{N}$ such that $\mathbf{E}[X^n] < \infty$ is $n = 3$. Indeed,

$$\mathbf{E}[(1 + X)^3] = \int_{-\infty}^{\infty} (1 + x)^3 f_X(x) dx = \int_0^{\infty} \frac{4}{(1 + x)^2} dx = \left[-\frac{4}{1 + x} \right]_0^{\infty} = 4,$$

and clearly $\mathbf{E}[X^3] \leq \mathbf{E}[(1 + X)^3]$. Moreover, $\mathbf{E}[X^4]$ does not exist: Indeed,

$$\mathbf{E}[X^4] = \int_0^{\infty} \frac{4x^4}{(1 + x)^5} dx \geq \int_1^{\infty} \frac{4x^4}{(2x)^5} dx = \frac{1}{8} \int_1^{\infty} \frac{1}{x} dx = \infty,$$

where we used that for $x \geq 1$, $1 + x \leq 2x$.

(b) i. We see that $f_X \geq 0$. Now,

$$\begin{aligned} \int_{-\infty}^{\infty} f_X(x) dx &= \int_0^{\infty} \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x} dx = \frac{1}{\Gamma(\alpha)} \int_0^{\infty} (\lambda x)^{\alpha-1} e^{-\lambda x} \lambda dx \\ &= \frac{1}{\Gamma(\alpha)} \int_0^{\infty} y^{\alpha-1} e^{-y} dy = 1, \end{aligned}$$

where we used the definition of the Γ -function and the substitution $y = \lambda x$.

ii. We write

$$\begin{aligned} \Gamma(\alpha + 1) &= \int_0^{\infty} x^\alpha e^{-x} dx \\ &= \underbrace{[-x^\alpha e^{-x}]_0^{\infty}}_{=0} + \int_0^{\infty} \alpha x^{\alpha-1} e^{-x} dx = \alpha \Gamma(\alpha). \end{aligned}$$

Here we used that $\lim_{x \rightarrow \infty} \frac{x^\alpha}{e^x} = 0$ for every $\alpha > 0$. Note that

$$\Gamma(1) = \int_0^{\infty} e^{-x} dx = [-e^{-x}]_0^{\infty} = 1,$$

so we see that

$$\Gamma(n) = (n - 1)\Gamma(n - 1) = (n - 1)(n - 2)\Gamma(n - 2) = \dots = (n - 1)!\Gamma(1) = (n - 1)!.$$

iii. We calculate

$$\begin{aligned} \mathbf{E}[X] &= \int_{-\infty}^{\infty} x f_X(x) dx = \frac{1}{\Gamma(\alpha)} \int_0^{\infty} \lambda^\alpha x^\alpha e^{-\lambda x} dx \\ &= \frac{\Gamma(\alpha + 1)}{\lambda \Gamma(\alpha)} \cdot \underbrace{\int_0^{\infty} \frac{1}{\Gamma(\alpha + 1)} \lambda^{\alpha+1} x^\alpha e^{-\lambda x} dx}_{=1} = \frac{\alpha \Gamma(\alpha)}{\lambda \Gamma(\alpha)} = \frac{\alpha}{\lambda}. \end{aligned}$$

iv. In order to determine the variance, we first calculate $\mathbf{E}[X^2]$:

$$\begin{aligned} \mathbf{E}[X^2] &= \int_{-\infty}^{\infty} x^2 f_X(x) dx = \frac{1}{\Gamma(\alpha)} \int_0^{\infty} \lambda^\alpha x^{\alpha+1} e^{-\lambda x} dx \\ &= \frac{\Gamma(\alpha + 2)}{\lambda^2 \Gamma(\alpha + 2)} \cdot \underbrace{\int_0^{\infty} \frac{1}{\Gamma(\alpha + 2)} \lambda^{\alpha+2} x^{\alpha+1} e^{-\lambda x} dx}_{=1} = \frac{(\alpha + 1)\alpha \Gamma(\alpha)}{\lambda^2 \Gamma(\alpha)} = \frac{(\alpha + 1)\alpha}{\lambda^2}. \end{aligned}$$

So:

$$\text{Var}[X] = \mathbf{E}[X^2] - \mathbf{E}[X]^2 = \frac{(\alpha + 1)\alpha}{\lambda^2} - \frac{\alpha^2}{\lambda^2} = \frac{\alpha}{\lambda^2}.$$

2. Properties of the expectation

Let $(\Omega, \mathcal{F}, \mathbf{P})$ be a probability space.

- (a) Let $X : (\Omega, \mathcal{F}) \rightarrow (\mathbb{N}, \mathcal{P}(\mathbb{N}))$ be a (discrete) $\mathbb{N}$ -valued random variable. Show that

$$\mathbf{E}[X] = \sum_{n=1}^{\infty} \mathbf{P}[X \geq n],$$

where we assume that $\mathbf{E}[X] < \infty$. Use this to calculate $\mathbf{E}[X]$ if $X \sim \text{Geo}(p)$, $p \in (0, 1)$.

Hint: Explain first why for nonnegative numbers $(a_{n,k})_{n,k \in \mathbb{N}}$, one has that $\sum_{k=1}^{\infty} \sum_{n=1}^{\infty} a_{n,k} = \sum_{n=1}^{\infty} \sum_{k=1}^{\infty} a_{n,k}$.

- (b) If $X : (\Omega, \mathcal{F}) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ is a non-negative real random variable (i.e. $X(\omega) \geq 0$ for every $\omega \in \Omega$), then one can show similarly to the above formula that

$$\mathbf{E}[X] = \int_0^{\infty} (1 - F_X(x)) dx,$$

where we assume that $\mathbf{E}[X] < \infty$. Use this formula to calculate $\mathbf{E}[X]$ for $X \sim \mathcal{E}(\lambda)$ for $\lambda > 0$.

- (a) For an $\mathbb{N}$ -valued random variable X we have

$$\begin{aligned} \mathbf{E}[X] &= \sum_{n=1}^{\infty} n \mathbf{P}[X = n] = \sum_{n=1}^{\infty} \sum_{k=1}^n \mathbf{P}[X = n] = \sum_{n=1}^{\infty} \sum_{k=1}^{\infty} \mathbf{P}[X = n] \mathbb{1}_{\{k \leq n\}} \\ &= \sum_{k=1}^{\infty} \sum_{n=1}^{\infty} \mathbf{P}[X = n] \mathbb{1}_{\{k \leq n\}} = \sum_{k=1}^{\infty} \sum_{n=k}^{\infty} \mathbf{P}[X = n] = \sum_{k=1}^{\infty} \mathbf{P}[X \geq k]. \end{aligned}$$

Now for a geometrically distributed random variable $X \sim \text{Geo}(p)$ and

$$\mathbf{P}[X = k] = p(1-p)^{k-1}, \quad k \in \mathbb{N} = \{1, 2, 3, \dots\}.$$

It follows that

$$\mathbf{P}[X \leq n-1] = \sum_{k=1}^{n-1} \mathbf{P}[X = k] = p \sum_{k=0}^{n-2} (1-p)^k = p \frac{1 - (1-p)^{n-1}}{1 - (1-p)} = 1 - (1-p)^{n-1}.$$

So:

$$\mathbf{P}[X \geq n] = 1 - \mathbf{P}[X \leq n-1] = (1-p)^{n-1}.$$

We obtain that

$$\mathbf{E}[X] = \sum_{n=1}^{\infty} \mathbf{P}[X \geq n] = \sum_{n=1}^{\infty} (1-p)^{n-1} \sum_{n=0}^{\infty} (1-p)^n = \frac{1}{1 - (1-p)} = \frac{1}{p}.$$

- (b) The cumulative distribution function of $X \sim \mathcal{E}(\lambda)$ is given by

$$F_X(x) = 1 - e^{-\lambda x}, \quad x \geq 0.$$

It follows that

$$\mathbf{E}[X] = \int_0^{\infty} (1 - F_X(x)) dx = \int_0^{\infty} e^{-\lambda x} dx = \left[-\frac{1}{\lambda} e^{-\lambda x} \right]_0^{\infty} = \frac{1}{\lambda}.$$

3. (*) Hazard rate

[4 Points]

We model the life-time of a device by a continuously distributed random variable $T \geq 0$ with cumulative distribution function $F(t)$ and probability density function $f(t)$. We assume that f is continuous in every $t \geq 0$. The *hazard rate* is defined as

$$\lambda(t) = \lim_{h \searrow 0} \frac{\mathbf{P}[t \leq T \leq t+h | T \geq t]}{h}, \quad t \geq 0$$

(this is also relevant for life-insurance purposes).

- (a) Express $\lambda(t)$ in terms of $f(t)$ and $F(t)$.
(b) Suppose that $T \sim \mathcal{E}(\alpha)$ with $\alpha > 0$, and calculate $\lambda(t)$.
(c) Verify that $F(t) = 1 - \exp\left(-\int_0^t \lambda(s) ds\right)$, when $t \geq 0$.
(d) Calculate $F(t)$ and $f(t)$ for $t \geq 0$ when $\lambda(t) = \alpha t^\gamma$ ($\alpha > 0, \gamma > 0$) (Weibull distribution).
(e) For $T \geq 0$, show that $\mathbf{E}[T] = \int_0^\infty \exp\left(-\int_0^t \lambda(s) ds\right) dt$.

(a) We have

$$\begin{aligned} \frac{\mathbf{P}[t \leq T \leq t+h \mid T \geq t]}{h} &= \frac{\mathbf{P}[\{t \leq T \leq t+h\} \cap \{T \geq t\}]}{\mathbf{P}[T \geq t] \cdot h} = \frac{\mathbf{P}[t \leq T \leq t+h]}{\mathbf{P}[T \geq t] \cdot h} \\ &= \frac{F(t+h) - F(t)}{h} \cdot \frac{1}{1 - F(t)} \\ &\rightarrow \frac{f(t)}{1 - F(t)} \quad (h \searrow 0). \end{aligned}$$

Thus, we have

$$\lambda(t) = \frac{f(t)}{1 - F(t)}.$$

(b) In this case, we have

$$f(t) = \begin{cases} 0 & \text{if } t < 0 \\ \alpha \exp(-\alpha t) & \text{if } t \geq 0 \end{cases} \quad \text{and} \quad F(t) = \begin{cases} 0 & \text{if } t < 0 \\ 1 - \exp(-\alpha t) & \text{if } t \geq 0 \end{cases}$$

so $\lambda(t) = \alpha$ for all $t \geq 0$.

(c) According to (a), we have

$$\begin{aligned} 1 - \exp\left(-\int_0^t \lambda(s) ds\right) &= 1 - \exp\left(-\int_0^t F'(s)/(1 - F(s)) ds\right) \\ &= 1 - \exp\left([\log(1 - F(s))]_0^t\right) \\ &= 1 - \exp(\log(1 - F(t)) - \log 1) \\ &= 1 - (1 - F(t)) = F(t). \end{aligned}$$

(d) According to (c) we have

$$\begin{aligned} F(t) &= 1 - \exp\left(-\int_0^t \alpha s^\gamma ds\right) = 1 - \exp\left(-\left[\frac{\alpha}{\gamma+1} s^{\gamma+1}\right]_0^t\right) \\ &= 1 - \exp\left(-\frac{\alpha}{\gamma+1} t^{\gamma+1}\right) \end{aligned}$$

and therefore

$$f(t) = \frac{d}{dt} F(t) = \alpha t^\gamma \exp\left(-\frac{\alpha}{\gamma+1} t^{\gamma+1}\right), \quad t \geq 0.$$

(e) We apply Problem 2(b) and (c) of this problem together:

$$\begin{aligned} \mathbf{E}[T] &= \int_0^\infty (1 - F(t)) dt = \int_0^\infty \left(1 - \left(1 - \exp\left(-\int_0^t \lambda(s) ds\right)\right)\right) dt \\ &= \int_0^\infty \exp\left(-\int_0^t \lambda(s) ds\right) dt \end{aligned}$$

4. (*) **More on expectation**

[4 points]

- (a) Let $\mathcal{S}_n$ be set of all permutations of $\{1, \dots, n\}$ and $\mathbf{P} = \mathcal{U}(\mathcal{S}_n)$ be the uniform distribution on $\mathcal{S}_n$. For every permutation $\omega \in \mathcal{S}_n$, let $X(\omega)$ be the number of fixed points of ω . Find $\mathbf{E}[X]$.
Hint: Consider for $i = 1, \dots, n$, $A_i = \{\omega \in \mathcal{S}_n : \omega(i) = i\}$. Then $X = \sum_{1 \leq i \leq n} \mathbf{1}_{A_i}$.
- (b) In a fictional company, a worker's legal code specifies as a holiday any day during which at least one worker has their birthday. All other days are working days. How many workers must the factory employ so that the expected number of working worker-days is maximized during the year? (Worker-days = Headcount $\times$ Days.)
Hint: First consider the expectation of the total number of birthday-free days.

- (a) Note that $\mathbf{1}_{A_i}$ records whether i is the fixed point of ω . Hence X , the total number of fixed points of ω , equals $\sum_{1 \leq i \leq n} \mathbf{1}_{A_i}$. As $\mathbf{P}$ is the uniform distribution,

$$\mathbf{P}[A_i] = \frac{(n-1)!}{n!} = \frac{1}{n}.$$

Hence

$$\mathbf{E}[X] = \mathbf{E}\left[\sum_{i=1}^n \mathbf{1}_{A_i}\right] = \sum_{i=1}^n \mathbf{P}[A_i] = 1.$$

- (b) Let $f(n)$ be the expected number of worker-days when n workers are employed. We seek to maximize f as a function of n . Thus, first we need to derive an explicit expression for f .

Remark: Even before we do so, it is easy to see that there must be at least one maximum for the function f . Clearly, $f(1) = 364$, and neglecting the probability of a birthday coincidence, $f(2) \approx 2 \cdot 363 = 726$. Thus, initially f increases. On the other hand, as $n \rightarrow \infty$, there will be no birthday-free day at all, meaning that the expected number of worker-days tends to zero as n gets very large. Together, then, f must have at least one maximum.

To locate this maximum, consider first some specific day i , where $i = 1, \dots, 365$. Let X_i be a random variable that equals 1 if none of the n workers has their birthday on this specific day whereas $X_i = 0$ if at least one worker has their birthday on that day. For any individual worker, the probability not to have their birthday on day i equals $364/365$. Therefore, with independent birthdays, the probability $\mathbf{P}[X_i = 1]$ that no worker has their birthday on day i is equal to $(364/365)^n$. This implies that the expectation $\mathbf{E}[X_i] = \mathbf{P}[X_i = 1] = (364/365)^n$.

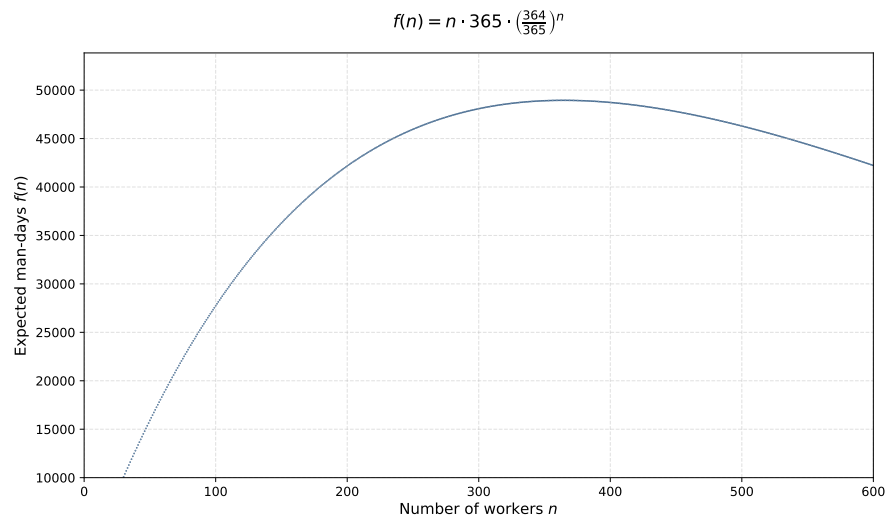
Now, define the random variable $S = \sum_{i=1}^{365} X_i$. In words, S is the total number of birthday-free days. Clearly, the X_i are not independent, but for the expectation $\mathbf{E}[S]$ we have

$$\mathbf{E}[S] = \mathbf{E}\left[\sum_{i=1}^{365} X_i\right] = \sum_{i=1}^{365} \mathbf{E}[X_i] = 365 \cdot (364/365)^n$$

Now, $f(n) = n \cdot \mathbf{E}[S]$, because in expectation the n workers will have to work on $\mathbf{E}[S]$ days, yielding $n \cdot \mathbf{E}[S]$ worker-days. Thus,

$$f(n) = n \cdot 365 \cdot (364/365)^n$$

The following Figure shows the function f .



Looking at the change of the sign of the difference $f(n+1) - f(n)$, it is easy to see that its maxima occurs at $n = 364$ and $n = 365$, yielding $f(364) = f(365) = 48943.5$ expected worker-days.